

Smart Kumbh Mela Safety & Assistance Platform

System Architecture & Build Reference

Final Year Project — Dept. of Computer Engineering
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1. Tech Stack

Layer	Technology
Frontend	React.js + Tailwind CSS — single app, role-based views (visitor / volunteer / admin / super admin)
Backend	Node.js + Express.js + Socket.IO — REST APIs, auth, real-time push
AI Service	Python + FastAPI + YOLOv8 (Ultralytics) + OpenCV + face-matching library (face_recognition / DeepFace)
Database	MongoDB + Mongoose (MongoDB Atlas, free tier)
Realtime transport	Socket.IO (backend ↔ frontend) · WebSocket (backend ↔ AI service)
Auth	JWT + bcrypt
Notifications	Twilio — SMS + Voice Call API
Maps / Routing	Leaflet.js + OpenStreetMap · Leaflet Routing Machine + OSRM (facility routing)
Hosting (demo)	Hostinger — frontend + Node backend · MongoDB Atlas — DB · Local GPU machine — AI service

2. Hardware & Software Requirements

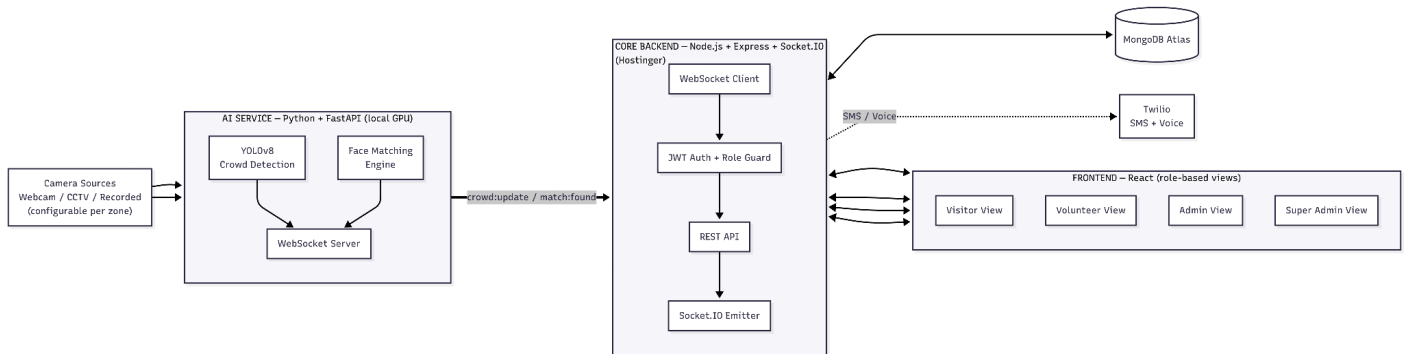
2.1 Hardware

Item	Purpose
Local system with a good GPU	Runs the Python AI service locally with GPU-accelerated YOLOv8 + face matching, for both build and live demo
Webcam or CCTV feed	Live crowd-monitoring demo, alongside the recorded sample video
QR wristbands / printed tags + printer	Physical ID issuance at checkpoint, mainly for kids/elderly/at-risk visitors
Smartphone(s)	Demo visitor app, SOS trigger, on-demand GPS for facility routing
Stable internet at demo location	WebSocket link between the local AI service and the Hostinger-deployed backend

2.2 Software & Tools

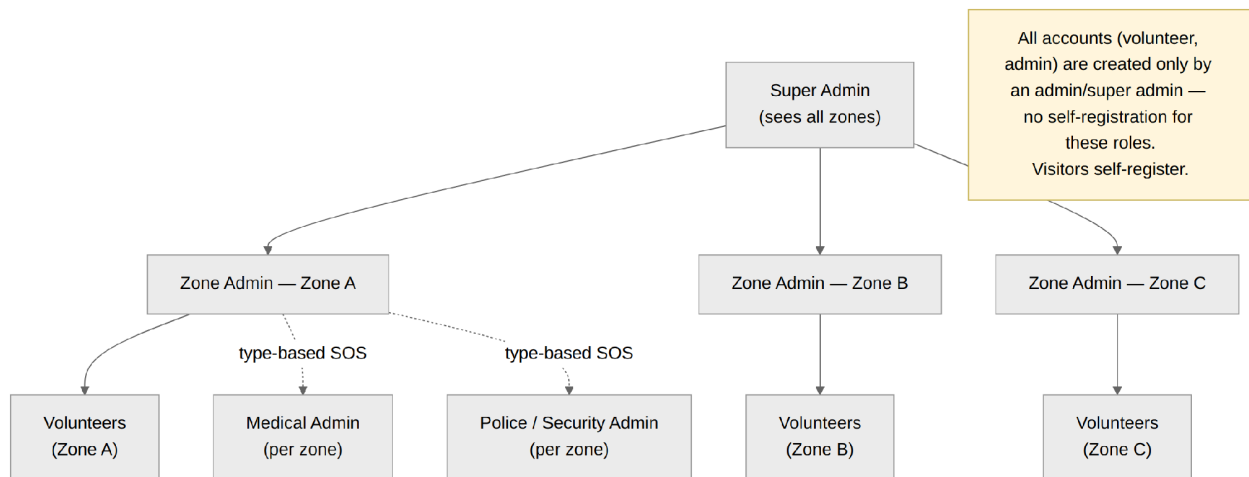
Tool / Service	Used for
VS Code	Primary IDE
Git + GitHub	Version control, team collaboration, PR reviews
Postman	API testing
MongoDB Compass	Inspecting Atlas collections during development
Python 3.10+	AI service runtime
Ultralytics YOLOv8 (pretrained weights)	Crowd/person detection — no training from scratch needed
face_recognition / DeepFace	Face-matching library for lost-person cases
Twilio account (SMS + Voice API keys)	SOS calls/texts and officer alerts
Leaflet.js + Leaflet Routing Machine + OSRM	Maps and facility routing
Hostinger account	Frontend + backend hosting
MongoDB Atlas account (free tier)	Hosted database

3. High-Level System Architecture

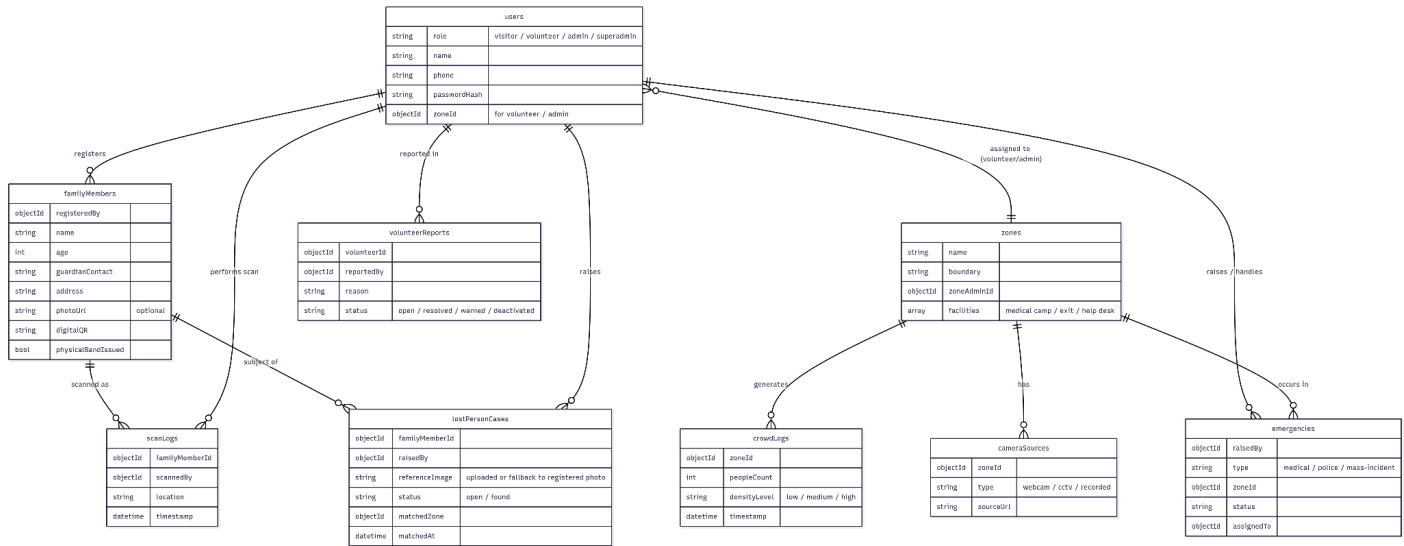


Three independent services: React frontend, Node.js core backend, Python AI service — connected via REST, Socket.IO, and WebSocket.

3.1 Roles & Zone Hierarchy

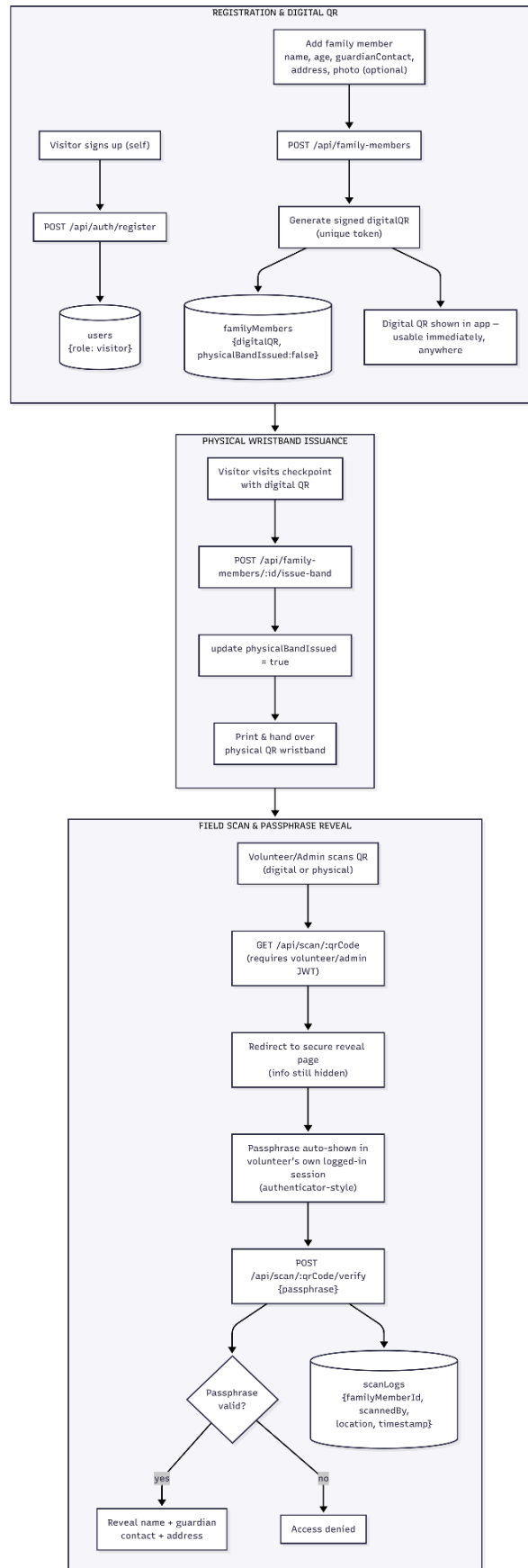


3.2 Data Model



Core MongoDB collections and their relationships.

4. Visitor & Family Registration + QR



- Any user registers self + family members from anywhere — not restricted to a checkpoint.
- Registration generates a digital QR per family member immediately; physical wristband QR is issued later at a checkpoint.
- Wristband is focused on kids/elderly/at-risk visitors but open to anyone.
- QR reveal is passphrase-gated (authenticator-style, shown in the scanning volunteer's own account) — not a shared secret.
- Every scan is logged centrally: location, scanning volunteer, timestamp, whose QR.

Scope: MVP

Endpoints: POST /api/auth/register · POST /api/family-members · POST /api/family-members/:id/issue-band · GET /api/scan/:qrCode · POST /api/scan/:qrCode/verify

Collections: users, familyMembers, scanLogs

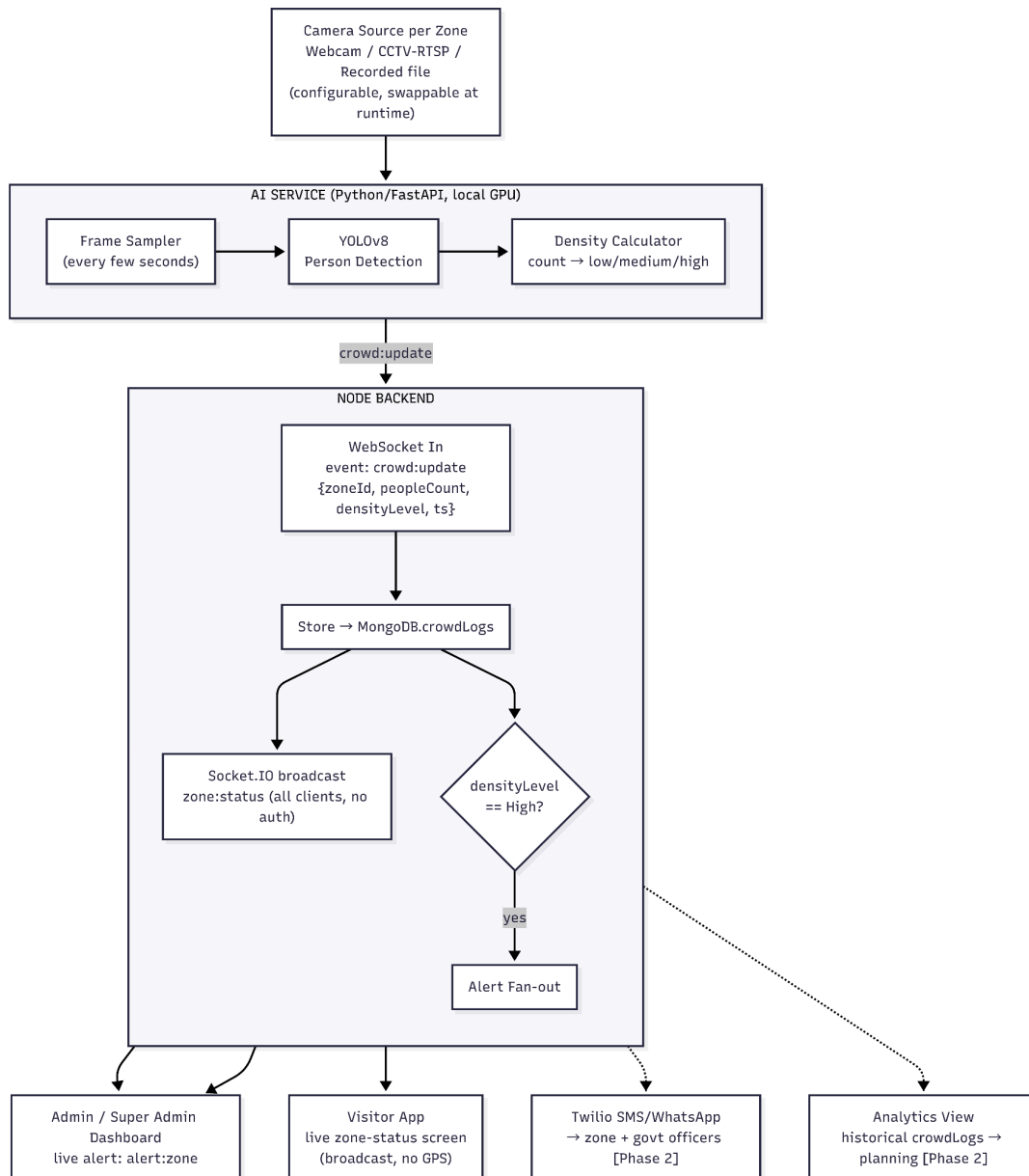
5. Crowd Monitoring

- Camera source is configurable per zone — webcam, CCTV/RTSP, or a recorded file — swappable without a code change.
- YOLOv8 computes people-count and density (low/medium/high) per zone, pushed to the backend over WebSocket.
- Dashboard alert is the baseline. Zone/government officer SMS-WhatsApp alerts and visitor rerouting are Phase 2.
- Live zone status is broadcast to all visitor app users (no location tracking, no permissions needed).
- Historical crowdLogs feed a planning/statistics view (Phase 2).

Scope: MVP core (dashboard alert + live broadcast) · Phase 2 (SMS/WhatsApp officer alerts, analytics)

Events: crowd:update (AI → backend) · zone:status (backend → all clients) · alert:zone (backend → admins)

Collections: crowdLogs, cameraSources, zones



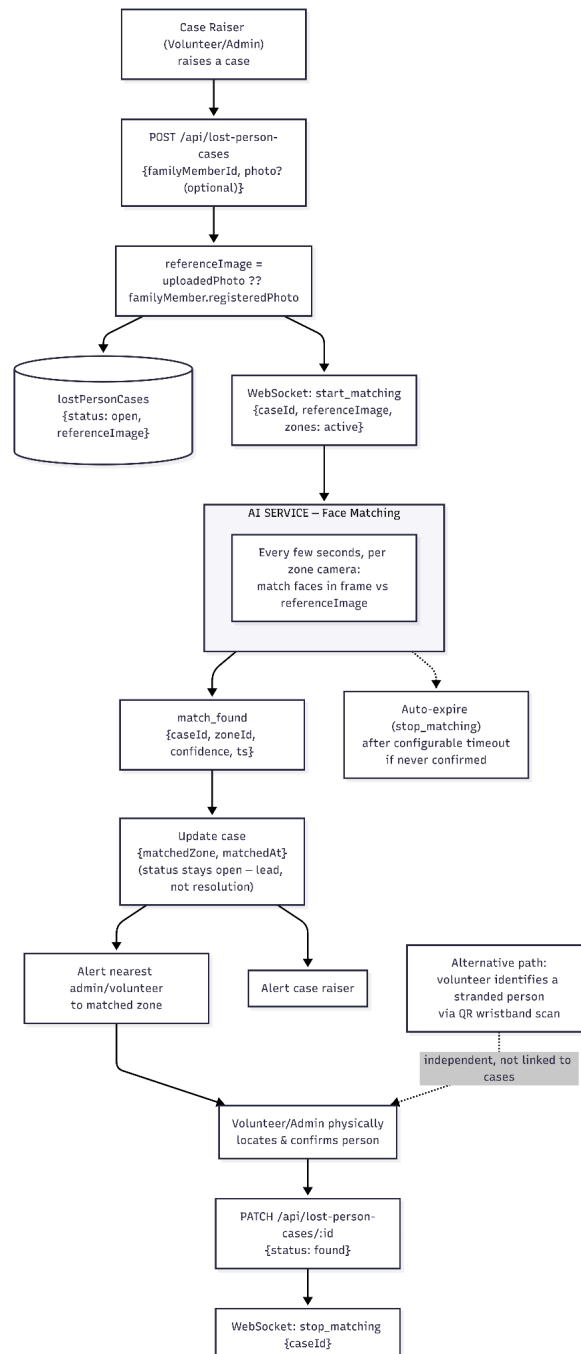
6. Lost Person Detection (Face Matching)

- Face matching runs only after a case is explicitly raised — not continuous — to avoid hardware overload.
- Reference photo: uploaded at case-raise time, falling back to the registered photo if one exists. No photo is mandated at registration.
- On match: nearest admin/volunteer to that camera's zone is alerted, plus whoever raised the case.
- A match is a lead, not automatic resolution — status stays "open" until a volunteer/admin confirms in person.
- QR wristband scan is a separate, independent identification path — intentionally not linked to face-matching cases.
- Case auto-expires after a configurable timeout if never confirmed.

Scope: MVP

Events: start_matching / stop_matching (backend → AI) · match_found (AI → backend)

Collections: lostPersonCases, familyMembers



7. SOS / Emergency

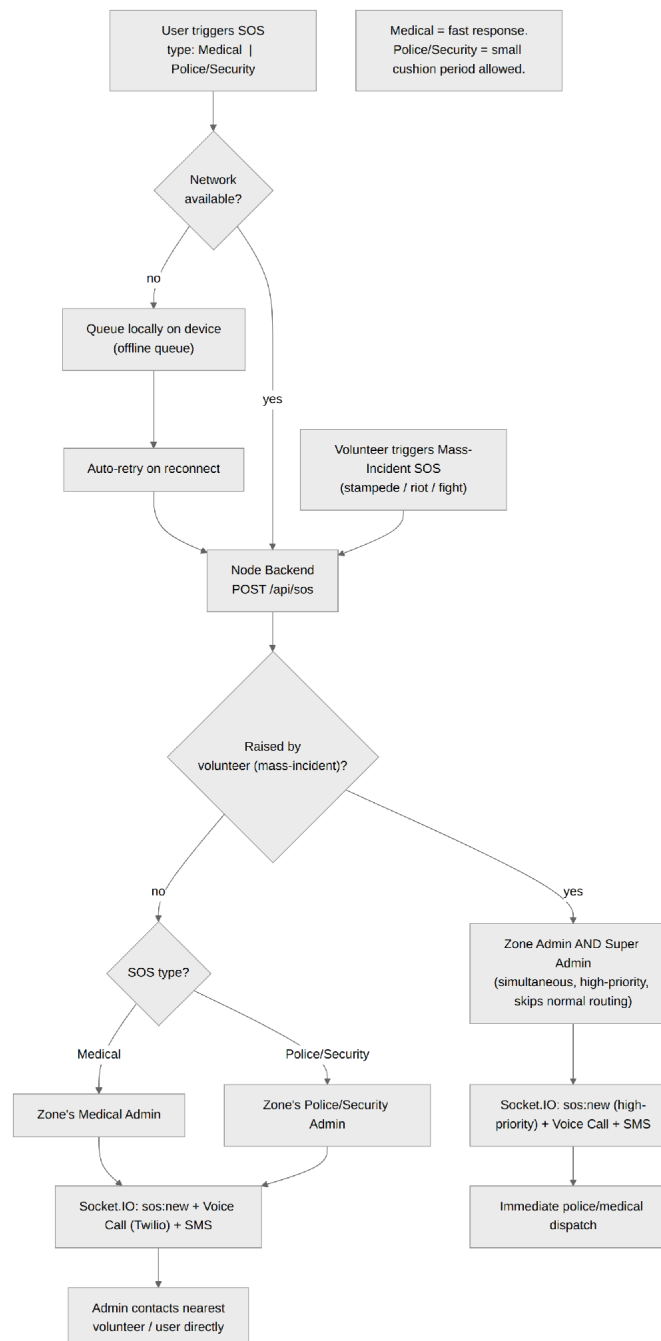
- Users raise Medical or Police/Security SOS — routed to that zone's type-specific admin, not straight to police/medical.
- Medical response is fast; Police/Security allows a small cushion period.
- Alerting includes an actual Twilio voice call + SMS, not just a dashboard/notification alert.
- Volunteer-raised mass-incident SOS (stampede/riot/fight) skips normal routing — goes straight to zone admin + super admin, high-priority.
- Poor-connectivity handling: SOS queues locally on the device and auto-retries once the connection returns.

Scope: MVP

Endpoints: POST /api/sos · PATCH /api/sos/:id/assign

Events: sos:new (backend → admin) · Twilio Voice Call + SMS

Collections: emergencies



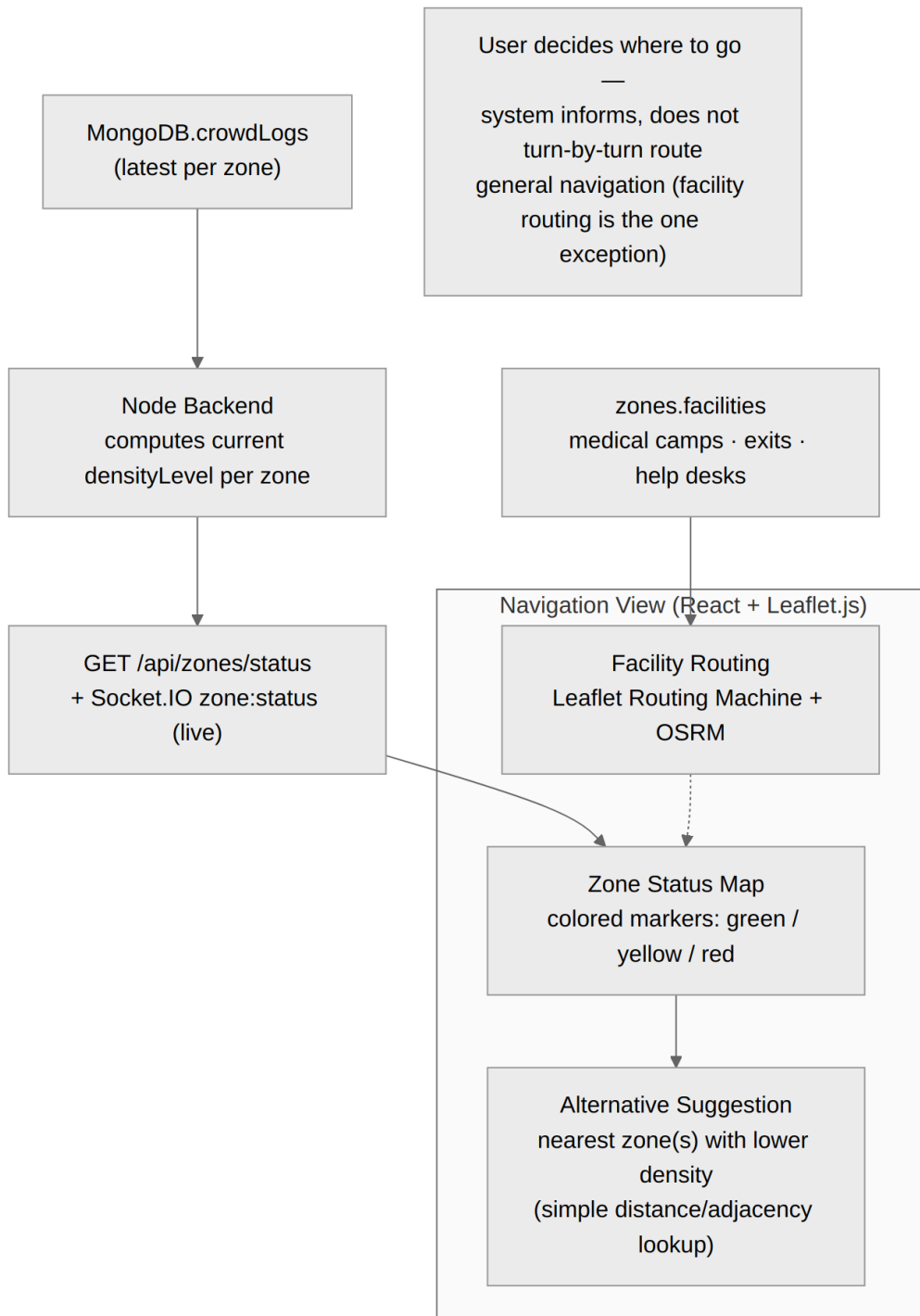
8. Navigation

- Not turn-by-turn in general — a crowd-awareness layer: shows which zone is overcrowded and less-crowded alternatives; the user decides where to go.
- Exception: fixed facility destinations (medical camps, exits, help desks) get real map/route-following guidance.
- Alternative-zone suggestion uses simple distance/adjacency lookup, not pathfinding.

Scope: MVP (moved up from stretch)

Endpoints: GET /api/zones/status (+ live via zone:status)

Collections: crowdLogs, zones.facilities



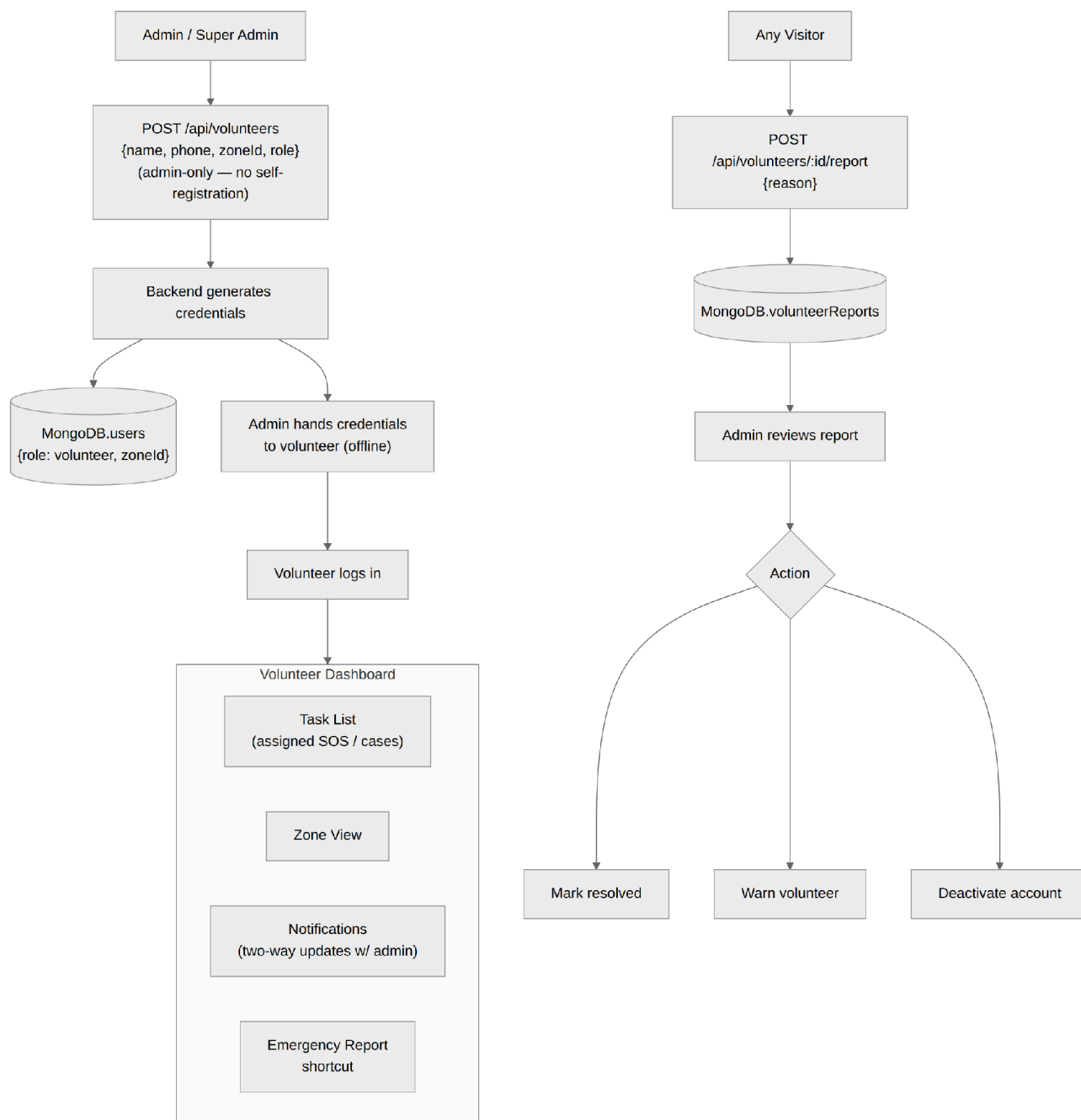
9. Volunteer Management

- Onboarding is admin-controlled only — an admin/super admin creates the account, assigns zone + role, and issues credentials. No self-registration for volunteers.
- Volunteers get their own account: task list, zone view, notifications (two-way with admin), emergency-report shortcut.
- Misconduct reporting: any visitor can report a volunteer by ID; admin can resolve, warn, or deactivate (Phase 2).

Scope: MVP (account creation, assignment, task view) · Phase 2 (misconduct reporting workflow)

Endpoints: POST /api/volunteers · GET /api/volunteers/tasks · POST /api/volunteers/:id/report

Collections: users (role: volunteer), volunteerReports



10. MVP (Phase 1) vs Phase 2

Module	MVP (Phase 1)	Phase 2 (stretch)
Registration + QR	Digital + physical QR, checkpoint issuance	—
Crowd Monitoring	Webcam/recorded input, dashboard alert, live broadcast, facility data	SMS/WhatsApp officer alerts · analytics/planning view
Lost Person	Case-based face matching, QR identification	—
SOS	Type-based routing, voice call + SMS, offline queue	—
Navigation	Zone-status alternatives + facility routing	—
Volunteer Mgmt	Admin-created accounts, zone assignment, task view	Misconduct reporting workflow

11. Deployment

- Frontend + Node backend: deployed on Hostinger.
- Database: MongoDB Atlas (free tier).
- AI service: runs locally on demo hardware (i7-13700HX, RTX 5060 8GB GPU, 16GB RAM) for GPU-accelerated inference, connecting to the deployed backend over WebSocket.
- This mirrors a realistic edge-inference pattern (local GPU inference + cloud-hosted control plane) rather than a budget workaround.

